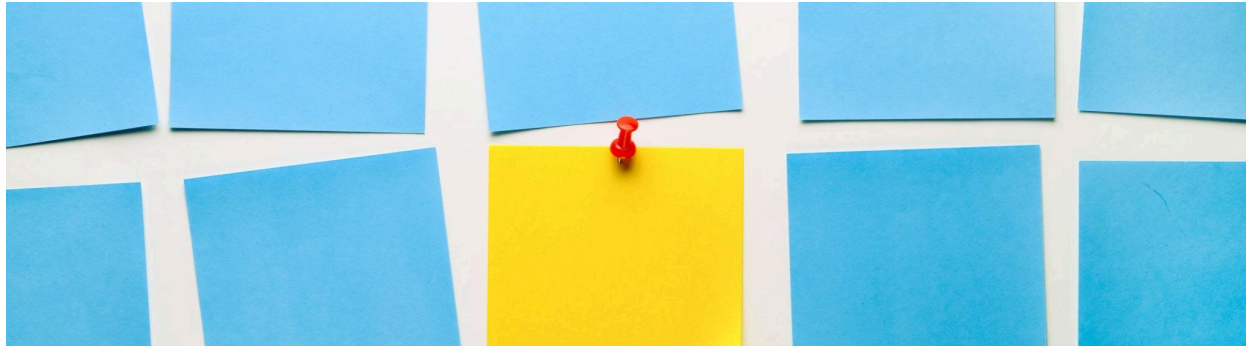




Business Requirements Document

Project Title: Improving Voice Input Adoption in ChatGPT Mobile App – India

Executive Summary

ChatGPT's mobile voice chat feature in India is witnessing lower-than-expected adoption, particularly among free-tier users. This document analyzes the core issues behind this trend, which include confusing microphone UI, unclear distinction between voice mode and speech-to-text, limited awareness of the feature's capabilities, especially in Tier 2 and Tier 3 cities, and barriers

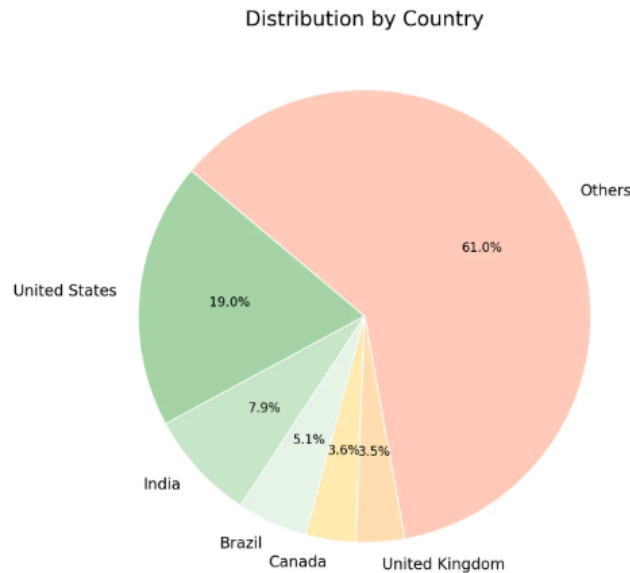
faced by non-tech-savvy users. Moreover, regional language clarity and trust in voice-based AI interactions continue to be important hurdles.

Through structured analysis of user segments, behavioral feedback (via app store reviews and social platforms), this Business Requirements Document (BRD) outlines the current and proposed user experience, business drivers, and both functional and non-functional requirements. The goal is to identify areas for improving discoverability, usability, and inclusiveness of the voice chat feature—especially during first-time use—without pre-defining final product features.

Project Description

ChatGPT launched voice chat on mobile in **March–April 2023**, powered by GPT-4o capabilities (multimodal, real-time speech) [Wikipedia](#). India makes up **~8.7% of global ChatGPT users** and is one of its fastest-growing markets [NerdyNav](#). For voice assistants across Asia, **62%** of users accessed voice tech in the past six months; in India, **>50%** use it monthly. All users—free and paid—are currently using GPT-4o-based voice. Free users receive a limited daily preview of advanced voice via GPT-4o mini, after that it switches to standard voice [OpenAI](#). Yet, ChatGPT's voice feature remains underutilized.

Countries with Highest ChatGPT Users



Project Scope

In-Scope:

- Improving discoverability, adoption, and user engagement for the voice chat feature in the ChatGPT mobile applications (iOS & Android).
- Support for major languages, including English, Hindi, and regional accents across Tier 1 to Tier 3 cities in India.

Out-Of-Scope:

- Upgrading LLM architecture beyond voice UX improvements.
- Changes to non-mobile platforms (web/desktop).
- Infrastructure-level backend optimizations

Business Drivers

Driver	Description
Retention & Engagement	Voice chat—especially high-quality GPT-4o mode—encourages longer, hands-free interaction compared to text.
User Acquisition / Accessibility	Non-English and low-literacy users prefer speaking; voice lowers barriers for Tier-2/3 adoption.
Market Differentiation	ChatGPT tops voice experience charts in comparisons like CreatorEconomy, offering richer, more personable voice interaction than Gemini or Perplexity (wired.com / CreatorEconomy). Siri and Alexa remain limited to commands and short interactions.
Use-Case Expansion	Voice allows new scenarios—like recipes, translations, or multitasking—that text-based modes can't serve.

Current Process / State

No Onboarding

New users receive no in-app prompt or tutorial describing the voice feature. Consequently, many—especially those in non-elite tiers or less tech-savvy segments—remain unaware of the feature.

Regional Language Challenges

Voice recognition struggles notably with Indian regional languages like Telugu and Tamil. A comparative study of language model competence showed that while ChatGPT handles Hindi relatively well, its Telugu proficiency lags significantly in grammar and vocabulary tasks, even compared to Gemini [arXiv](#). This results in instances where returns are incoherent or entirely in unrelated languages, as echoed in user anecdotes and app reviews. Recognizing this deficiency is essential to addressing trust and usability for regional users.

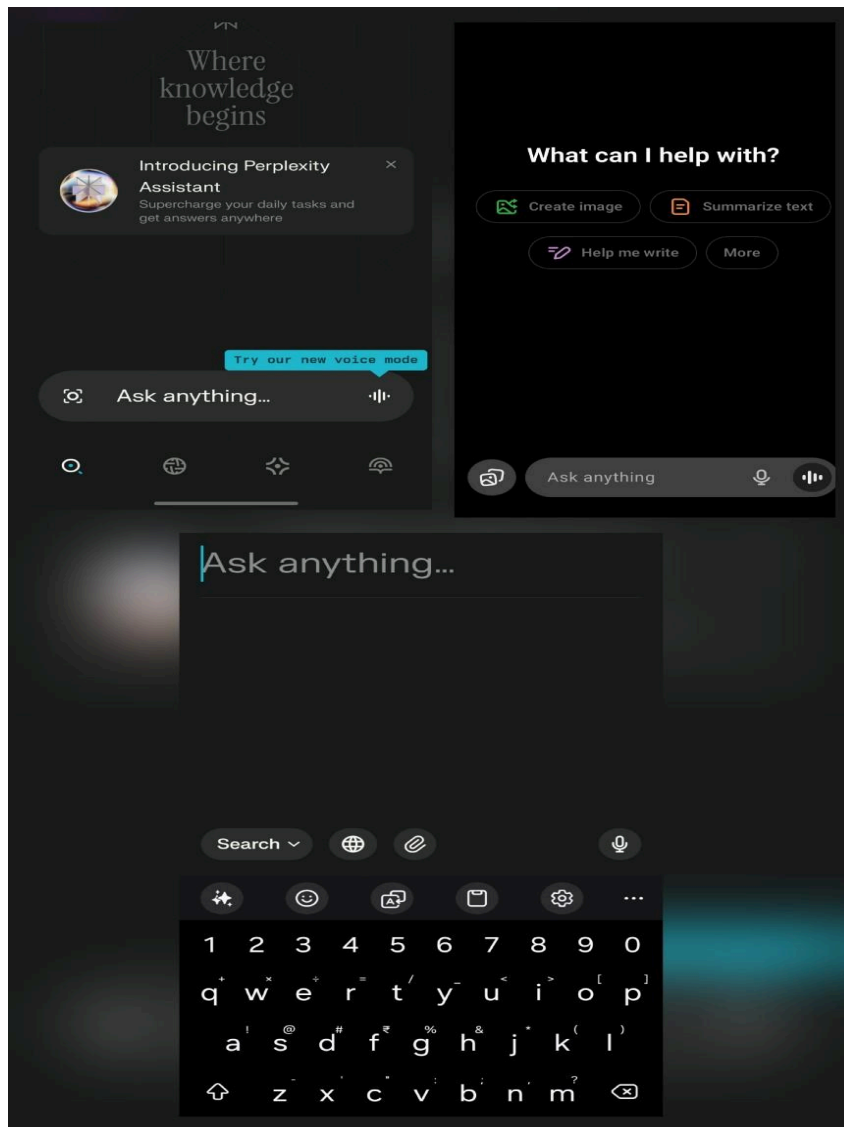
The BLEU score is a standard measure of machine translation quality, comparing machine-generated translations to a reference human translation on a scale of 0 to 100.

Language	BLEU Score
Hindi	72.69
Kannada	46.78
Telugu	13.12

Source: [arXiv](#)

Feature Misidentification

Due to limited clarity in UI presentation, users often mistake the microphone icon for basic speech-to-text functionality rather than the interactive voice-chat capability.



Competitor Advantage in Discovery

Competing apps like Perplexity and Gemini actively promote voice through onboarding pop-ups and immediate prompts. In contrast,

system assistants (e.g., Siri, Google Assistant) are preinstalled and become default for many users—all of which reinforce habitual voice usage. [MediumarXiv](#)

Proposed Process

To drive better clarity and increase voice feature adoption, especially among new and non-tech-savvy users, we propose the following improvements:

- **Voice onboarding prompt on first app launch:**
When a user opens the app for the first time, they should see a prompt introducing the voice feature—what it does, and how to use it—so that they’re aware of its existence from the start.
- **Mini walkthrough/tutorial for voice usage:**
A short, optional walkthrough should demonstrate how to activate the mic, distinguish between listening mode vs speech-to-text, and show examples of voice interactions to reduce confusion.
- **Smart nudges based on user behavior and locale:**
If a user interacts with the app through text for a few sessions, the app can gently suggest trying voice mode—tailored to their language and location, e.g., “Try asking in Hindi!” or “Want to speak instead of type?”

- **Regional language detection and voice tips:**
Automatically detect the user's regional language preference (e.g., based on device settings or prior usage) and offer relevant voice guidance—for example, showcasing that the app understands and responds in that language.
- **Improve regional language accuracy and BLEU score:**
Invest in enhancing speech recognition and response accuracy for major Indian languages. Better BLEU (Bilingual Evaluation Understudy) scores—used to evaluate translation quality, this can significantly boost user trust and satisfaction when engaging via voice in native languages.

These updates will make voice interactions more discoverable, intuitive, and localized—especially for users in Tier 2/3 cities, those unfamiliar with AI tools, and speakers of regional languages.

Functional Requirements

FR1: On app launch, first-time users should receive a modal explaining the voice chat capability.

FR2: A wave icon usable for voice chat must be persistently visible in non-text-entry contexts.

FR3: Users should be able to select or change voice/persona or language before using voice.

FR4: Voice sessions must support **multi-turn conversation**, preserving context over at least 2 minutes or ~2k tokens.

FR5: After early voice errors or misunderstanding, users must see a feedback prompt to tune voice preferences.

Non-Functional Requirements

NFR1: Audio latency must remain under **500ms**, even on 2G/3G networks in Tier 3 cities.

NFR2: System fallback via streaming or partial on-device recognition should handle unstable connectivity.

NFR3: Microphone access and voice data must be encrypted in transit and not retained or used for training without explicit user consent—consistent with privacy standards in India arxiv.org.

NFR4: Voice feature should not increase app binary size by more than 10MB or significantly drain battery (<5% per session average).

Assumptions & Constraints

Assumptions:

- Visible voice UI and efficient onboarding will meaningfully increase usage.
- Users across city tiers want to use voice if accessible.
- Voice recognition algorithms can adapt to regional Indian accents with tuning.

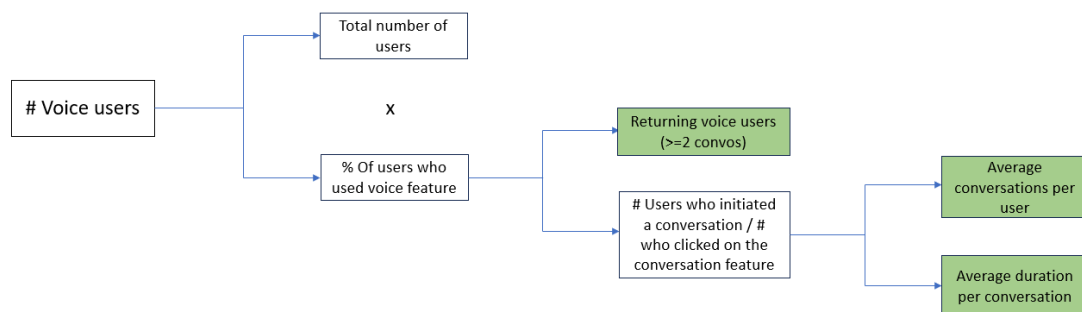
Constraints:

- Poor or patchy mobile data in Tier 2/3 areas.
- Accuracy limitations for lesser-used languages/dialects.

- On-device recording permissions and OS-level conflicts with default system assistants (like Siri/Assistant).

Success Criteria & KPIs

- 20% more users with ≥ 2 voice chats in 3 months.
- 15% increase in weekly voice sessions per user.
- Average voice chat > 2 minutes.
- Monitor **Voice Activation Rate**: (users who begin voice chat) $\div$ (users who tap mic icon).
- Increase positive voice mentions in app reviews by 25%.



Timeline & Milestones

Phase	Duration	Goals & Milestones
Discovery & Research	Weeks 1–2	Analyze analytics and user reviews; conduct rapid surveys (target: 100 voice-tech users).
Design & Prototyping	Weeks 3–5	Design onboarding and UI updates; usability test prototypes (target usability score: ≥80%).
Accessibility & Localization	Weeks 6–7	Localize voice prompts and personas; test Hindi/Hinglish recognition accuracy (~90%).
Pilot Rollout & A/B Test	Weeks 8–10	Launch updated UI to 20% of India users; measure activation conversion uplift (target: +10%).
Iteration & Full Launch	Weeks 11–13	Refine based on pilot metrics; full rollout; monitor KPI targets achieved (≥20% voice adoption growth).

Stakeholders

- Product Manager (owner)
- UX/UI Designer
- Mobile Engineers (iOS & Android)
- Localization / Language Engineers
- Data Analytics / Product Insights
- India GTM / Marketing
- Legal / Privacy & Compliance

Glossary

BRD (Business Requirements Document) – A formal document outlining business needs, goals, and functional/non-functional requirements for a project.

GPT-4o – OpenAI’s multimodal AI model capable of handling text, voice, and image input in real time.

GPT-4o mini – A lighter, faster variant of GPT-4o used for free-tier or low-latency scenarios.

LLM (Large Language Model) – An AI system trained on vast text datasets to understand and generate human-like language.

Multimodal – AI capability to process and respond using multiple input/output types (e.g., text, voice, image)

BLEU score (Bilingual Evaluation Understudy) – A metric (0–100) that measures the accuracy of machine-generated translations against human translations.

Tokens – Units of text (roughly words or word parts) used for measuring context length in language models.

Voice Activation Rate – The percentage of users who start a voice chat after seeing or tapping the microphone icon.

Activation conversion uplift – The improvement in user activation rate after implementing a change (e.g., new onboarding flow).

References

ChatGPT User Demographics: [explodingtopics.com](https://explodingtopics.com/blog/chatgpt-demographics)

Comparison Of Different AI Models: [Creatoreconomy](https://creatoreconomy.com/ai-model-comparison/)

ChatGPT Statistics: [nerdynav.com](https://nerdynav.com/chatgpt-statistics/)

GPT-4o Voice Capabilities: [wired.com](https://www.wired.com/story/gpt-4o-voice/)

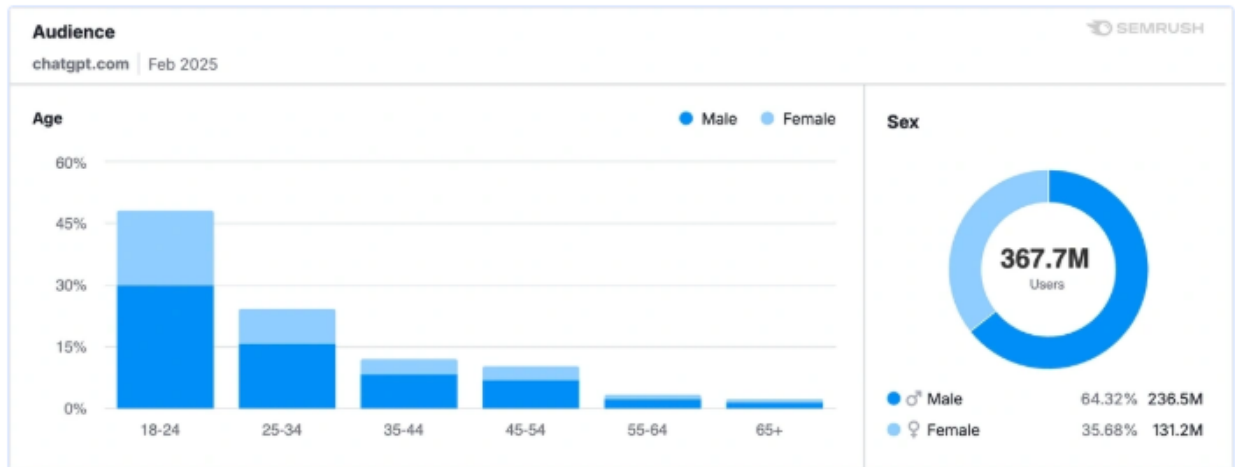
Comparing Multilingual Capabilities of ChatGPT: [ARXIV](https://arxiv.org/abs/2304.03472)

Appendix

ChatGPT User Demographics

64.32% of ChatGPT users are male, and 35.68% are female.

More than 45% of its users are under the age of 25.



Sources: [Semrush](#)

ChatGPT by Region

No country has more ChatGPT users than the US (16.18%).

Second-placed India (8.71%) has just over half as many ChatGPT users.

Together, the top five countries make up around 37% of all ChatGPT users.

Here's the breakdown of ChatGPT users by region:

Rank	Country	Proportion of ChatGPT Users
1	US	16.18%
2	India	8.71%
3	Brazil	5.28%
4	Germany	4.04%
5	Japan	2.9%
-	Others (combined)	62.89%

ChatGPT Traffic

Top Marketing Channels

Over 4 in 5 (81.38%) ChatGPT visits are direct.

Referral traffic is the second-most popular way to access ChatGPT (8.48% of traffic).

Here's a breakdown of the most popular ChatGPT marketing channels:

Rank	Marketing Channel	Proportion of Traffic
1	Direct	81.38%
2	Referral	8.48%
3	Organic Search	7.68%
4	Paid Search	1.27%
5	Organic Social	1.14%